

Operations research (20.02.2026)

All steps and computations must be clearly visible, don't do cutting or erasing and keep the work clean. Make new simplex table after completing every iteration.

Q1. Situation: Manufacturing of 3 products A, B and C.

Constraints: 1. Volume for A is at least 50% of the combined demand for the three products.

2. Max Demand for A = 120

3. Raw material availability, Max: 300 kg

Given, raw material consumption (**per unit of product**): 2kg for A, 4kg for B, 3kg for C.

Given, Unit selling price for A: Rs 20, B: Rs 50, C: Rs 30.

Determine:

1. Optimal product mix for the company using appropriate simplex method.
2. Dual price for raw material resource.
3. Individual feasibility ranges for the changes in R.H.S. of the Constraints
4. New Solution if max demand of A increases by 20 units

Marks breakup: 6+1+2+1 for part 1-4.